

HENRY R. SCHARF

CONTACT INFORMATION

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RESEARCH EMPLOYMENT

Assistant Professor 2023–Current
University of Arizona, Tucson, Arizona

Assistant Professor 2019–2023
San Diego State University, San Diego, California

Postdoctoral Fellow 2018–2019
Colorado State University, Fort Collins, Colorado

EDUCATION

Colorado State University, Fort Collins, Colorado USA

PhD in Statistics 2017

Advisor: Mevin Hooten

MS in Statistics 2014

University of Arizona, Tucson, Arizona USA

MEd in Teaching and Teacher Education 2008

BS in Mathematics and Physics 2007

Cum laude with Honors

PUBLICATIONS

Ellwanger, A. L., K. S. Morrow, A. K. Dhawale, **H. R. Scharf**, P. O. Ngakan, E. P. Riley (2026). Balancing risk and reward: Toward resilient human-primate coexistence in a rapidly changing environment in Sulawesi, Indonesia. *Ambio*, 55: 1127–1143.

Trinidad, J.*, **H. R. Scharf**, P. O. Ngakan, E. P. Riley (2025). Roadside dining: The collective movement behavior of Sulawesi Moor macaques in a provisioning context. *American Journal of Primatology*, 87: e23727.

Scharf, H. R., J. Schierbaum*, H. Matsumoto, T. Assal (2024). Predicting species-level vegetation cover using large satellite imagery data sets. *Journal of Agricultural, Biological and Environmental Statistics*, 1–20.

McFadden A. J.*, A. D. Stow, P. J. Riggan, R. Tissell, J. O’Leary, **H. R. Scharf** (2024). Estimating Fire Radiative Energy Density with Repeat-Pass Aerial Thermal-Infrared Imaging of Actively Progressing Wildfires. *Fire*, 7(6):179.

Williams, P. J., X. Lu, **H. R. Scharf**, M. B. Hooten (2023). Embracing asymmetry in nature: How to account for skewness in ecological data. *Ecological Informatics*, 75: 102085.

Boulil, Z. L.*, J. W. Durban, H. Fearnbach, T. W. Joyce, S. G. M. Leander, **H. R. Scharf** (2023). Detecting changes in dynamic social networks using multiply-labeled movement data. *Journal of Agricultural, Biological and Environmental Statistics*, 28: 243–259.

*Student contributor

Scharf, H. R. (2022). Local indicators of spatial association (LISA). *Wiley StatsRef: Statistics Reference Online*.

Scharf, H. R., X. Lu, P. J. Williams, M. B. Hooten (2022). Constructing flexible, identifiable and interpretable statistical models for binary data. *International Statistical Review*, 90: 328–345.

Raiho, A. M., **H. R. Scharf**, C. A. Roland, D. K. Swanson, S. E. Stehn, and M. B. Hooten (2022). Searching for refuge: A framework for identifying site factors conferring resistance to climate-driven vegetation change. *Diversity and Distributions*, 28(4), 793–809.

Scharf, H. R., A. M. Raiho, S. Pugh, C. A. Roland, D. K. Swanson, S. E. Stehn, M. B. Hooten (2022). Multivariate Bayesian clustering using covariate-informed components with application to boreal vegetation sensitivity. *Biometrics*, 78: 1427–1440.

Reimer, J. R., J. Arroyo-Esquivel, J. Jiang, **H. R. Scharf**, E. M. Wolkovich, K. Zhu, C. Boettiger (2021). Noise can create or erase long transient dynamics. *Theoretical Ecology*, 14: 685–695.

Scharf, H. R. (2021). Statistical analysis of animal movement: Understanding behavior through hierarchical parametric models. *Notices of the American Mathematical Society*, 68(6), 911–924.

Scharf, H. R., F. Buderman (2020). Animal movement models for multiple individuals. *Wiley Interdisciplinary Reviews: Computational Statistics*, e1506.

Scharf, H. R., M. B. Hooten, R. R. Wilson, G. M. Durner, T. C. Atwood (2019). Accounting for phenology in the analysis of animal movement. *Biometrics*, 75: 810–820.

Hooten, M. B., **H. R. Scharf**, J. M. Morales (2019). Running on empty: Recharge dynamics from animal movement data. *Ecology Letters*, 22, 377–389.

Hooten, M. B., **H. R. Scharf**, T. J. Hefley, A. T. Pearse, M. D. Weegman (2018). Animal movement models for migratory individuals and groups. *Methods in Ecology and Evolution*, 9, 1692–1705.

Scharf, H. R., M. B. Hooten, D. S. Johnson, J. W. Durban (2018). Process convolution approaches for modeling interacting trajectories. *Environmetrics*, e2487.

Scharf, H. R., M. B. Hooten, D. S. Johnson (2017). Imputation approaches for animal movement modeling. *Journal of Agricultural, Biological and Environmental Statistics*, 22(3), 335–352.

Hefley, T. J., K. M. Broms, B. M. Brost, F. E. Buderman, S. L. Kay, **H. R. Scharf**, J. R. Tipton, P. J. Williams, and M. B. Hooten. (2017). The basis function approach to modeling autocorrelation in ecological data. *Ecology*, 98(3), 632–646.

Scharf, H. R., M. B. Hooten, B. K. Fosdick, D. S. Johnson, J. M. London, and J. W. Durban. (2016). Dynamic social networks based on movement. *Annals of Applied Statistics*, 10(4), 2182–2202.

	<i>CAREER: Learning social structure from movement data</i> National Science Foundation, Div Of Biological Infrastructure. Total Award: \$644,716. Role: PI. Status: Unfunded. Revised submission planned for 2026 cycle.	2026-2031
	<i>Focused Next Steps for Behavioral and Physiological Response Studies with Social Delphinids off Southern California</i> Office of Naval Research, Marine Mammals and Biology. Primary Institution: Southall Environmental Associates, Inc. Sub-Award to UofA: \$130,000. Role: Senior personnel. Status: Terminated 2025-04-03 with \$48,000 remaining.	2024–2026
	<i>Factors affecting provisioning and foraging in rapidly changing landscapes</i> National Science Foundation, SBE. Total Award: \$329,849. Role: Co-PI. Status: Funded	2023–2027
	<i>Collaborative Research: Pacific Alliance for Low-Income Inclusion in Statistics & Data Science</i> National Science Foundation, S-STEM. Total Award: \$5,000,000. Role: Co-PI 2022–2023. Status: Funded	2022–2028
	<i>Visualizing trajectory uncertainty</i> University Grants Program, SDSU. Total Award: \$10,000. Role: PI. Status: Funded	2021–2022
	<i>Broadening the impact and accessibility of animal movement models</i> University Grants Program, SDSU. Total Award: \$10,000. Role: PI. Status: Funded	2020–2021
INVITED CONFERENCE PRESENTATIONS	<i>Predicting fine-scale taxonomic variation in landscape vegetation using large satellite imagery data sets</i> EnviBayes Workshop on Complex Environmental Data, Fort Collins, CO, USA	2023
	<i>Detecting changes in dynamic social networks using unlabeled movement data</i> Joint Statistical Meetings, Washington D.C., USA	2022
	<i>Detecting changes in dynamic social networks based on unlabeled movement data</i> CMStatistics, London, UK	2021
	<i>Multivariate Bayesian clustering using covariate-informed components with application to boreal vegetation sensitivity</i> Joint Statistical Meetings, Seattle, WA, USA	2021

	<i>Statistical models for dependent trajectories with application to animal movement</i>	2019
	Joint Statistical Meetings, Denver, CO, USA Topic contributed session for ISBA Savage Award finalists	
	<i>Animal movement models for migratory individuals and groups</i>	2018
	The Wildlife Society Annual Conference, Cleveland, OH Symposium: <i>Animal movement: Advances in movement modeling and their applications</i>	
	<i>Imputation approaches for animal movement modeling</i>	2018
	Joint Statistical Meetings, Vancouver, BC, Canada Invited session	
	<i>Dynamic Social Networks Based on Movement</i>	2016
	Joint Statistical Meetings, Chicago, IL Student paper award winner (ENVR)	
DEPARTMENTAL SEMINARS	<i>A strategy for avoiding particle degeneracy in recursive Bayesian inference</i>	2024
	Department of Statistics, Pennsylvania State University	
	<i>Predicting species-level vegetation cover using large satellite imagery data sets</i>	2024
	Department of Ecosystems Science and Management, Pennsylvania State University	
	<i>Hierarchical statistical models can fuse multiple data sources and account for dependence in observational data</i>	2024
	School of Natural Resources and the Environment, University of Arizona	
	<i>Detecting changes in dynamic social networks based on unlabeled movement data</i>	2021
	Department of Mathematics and Statistics, California State University Long Beach	
	<i>Movement data reveal dynamic social relationships</i>	2021
	Department of Statistics, University of British Columbia	
	<i>Movement data reveal dynamic social relationships</i>	2021
	Department of Statistics, Kansas State University	
	<i>Multi-layered convolutional Gaussian process models for animal movement</i>	2020
	Computational Sciences Research Center, San Diego State University	
	<i>Statistical models for heterogeneous animal movement</i>	2019
	Department of Statistics & Data Science, University of Texas at Austin	
CONTRIBUTED PRESENTATIONS, COMPETITIVE SELECTION	<i>Wait, who was that? Tracking interacting trajectories with label uncertainty</i>	2026
	ISBA World Meeting, Nagoya, Japan	

<i>New tools for recursive Bayesian inference applied to high-resolution satellite imagery</i>	2025
Joint Statistical Meetings, Nashville, TN, USA	
<i>A strategy for avoiding particle degeneracy in recursive Bayesian inference</i>	2024
Joint Statistical Meetings, Portland, OR, USA	
<i>A strategy for avoiding particle degeneracy in recursive Bayesian inference</i>	2024
Western North American Region of The International Biometric Society annual meeting , Fort Collins, CO, USA	
<i>Identifying species-level vegetation cover using Sentinel-2 imagery</i>	2022
ASA Section on Statistics and the Environment Workshop, Provo, Utah	
<i>Accounting for phenology in the analysis of animal movement</i>	2018
International Statistical Ecology Conference, University of St. Andrews, Scotland	
<i>Process convolution approaches for modeling interacting trajectories</i>	2017
Joint Statistical Meetings, Baltimore, MD	
<i>Process convolution approaches for modeling interacting trajectories</i>	2017
Statistics Department at Colorado State University, Fort Collins, CO. Poster	
<i>Dynamic Social Networks Based on Movement</i>	2016
International Statistical Ecology Conference, Seattle, WA	
<i>Spatiotemporal Models for Animal Social Structure</i>	2015
Joint Statistical Meetings, Seattle, WA	
<i>Spatiotemporal Models for Animal Social Structure</i>	2015
Statistical and Applied Mathematical Sciences Institute ECOL: Transitional Workshop Research Triangle Park, NC	
<i>Dynamic social networks based on movement</i>	2015
Graduate Student Showcase, Fort Collins, CO. Poster	
<i>Spatiotemporal models for animal social structure</i>	2015
Statistics Department at Colorado State University, Fort Collins, CO. Poster	
<i>Novel visualization and analysis for extreme-scale wind turbine simulations</i>	2014
Conference on Data Analysis, Santa Fe, NM. Poster	

TEACHING

<i>Intro. to Statistical Computing</i> (Arizona: DATA 375)	Spring 2026
<i>Environmental Statistics</i> (Arizona: MATH/STAT/BIOS 574E)	Fall 2024 Fall 2025

<i>Design of Experiments</i> (Arizona: MATH/STAT 571B)	Spring 2024 Spring 2025 Spring 2026
<i>Advanced Statistical Regression Analysis</i> (Arizona: MATH/STAT 571A)	Fall 2023
<i>R Programming and Data Science</i> (SDSU: STAT 410)	Spring 2023 Fall 2022
<i>Bayesian Statistics</i> (SDSU: STAT 676)	Spring 2022
<i>Applied Spatio-Temporal Statistics</i> (SDSU: STAT 596/696)	Spring 2023 Spring 2021 Spring 2020
<i>Statistical Computing</i> (SDSU: STAT 580)	Fall 2022 Fall 2021 Fall 2020
<i>Introduction to Statistical Learning</i> (SDSU: STAT 596)	Fall 2019

ADVISING & MENTORING

Current

<i>Postdoctoral scholars</i> Anisha Mathias	Expected Spring 2027
<i>PhD students</i> Alexander Berliner	Expected Spring 2027
<i>Masters students</i> Salma Khantoon	Expected Summer 2026
<i>Undergraduate students</i> Alex Shah	Expected Spring 2028

Past, University of Arizona

<i>Masters students</i> Josephine Ward	Summer 2025
<i>Undergraduate students</i> Ian Stewart	Spring 2025

Past, SDSU

<i>Masters students</i> Hugo Rosales Portillo	Summer 2023
Angelica Rivera	Spring 2023
Navid Nezamabadi	Summer 2022
Jonathan Schierbaum	Spring 2022
Zaineb Boulil	Spring 2021
Jennifer Betancourt	Spring 2021
Kristine Dinh	Spring 2021
<i>Undergraduate students</i> Elena Dubocanin	Winter 2022

WORKSHOPS & SHORT COURSES

Workshop: Statistical Models for Animal Movement

	Co-instructor Day-long workshop on animal movement modeling with examples in R International Association for Landscape Ecology annual meeting Fort Collins, CO, USA	2019
	International Statistical Ecology Conference University of St. Andrews, Scotland	2018
	<i>Training in Bayesian Modeling for Practicing Ecologists.</i> Co-instructor NSF-supported two-week workshop Intensive training in Bayesian modeling Colorado State University, Fort Collins, CO	2016
	<i>R Workshop</i> Co-instructor Day-long tutorial on R for ecologists Colorado State University, Fort Collins, CO	2015
	<i>Tutorial on Parallel Programming in R</i> Co-Developer Workshop on Parallel Computing, UC Boulder/CSU	2015
	<i>Tutorial on Parallel Programming in R</i> Co-Developer Conference on Statistical Practice 2015, New Orleans, LA	2015
AWARDS	ISBA Savage Award - Applied Methodology <i>The Savage Award, named in honor of Leonard J. "Jimmie" Savage, is bestowed each year to two outstanding doctoral dissertations in Bayesian econometrics and statistics.</i> Amount: \$750 + \$500 travel support	2018
	James L., M. Leslie, & Edna Madison Memorial Award <i>This award is given each year to the student selected by the statistics faculty at Colorado State University as the outstanding graduate student in the Department of Statistics and/or the Department of Mathematics</i> Amount: \$675	2017
SERVICE	Award Committee: ISBA Blackwell-Rosenbluth Award for outstanding junior Bayesian researchers	2025
	Scientific Committee: EnviBayes Workshop on Complex Environmental Data	2025
	Program Officer: International Society for Bayesian Analysis Section on Environmental Statistics (EnviBayes)	2023-2024
	Award Committee: ISBA Savage Award - Applied Methodology for outstanding dissertations in Bayesian econometrics and statistics	2021
	Secretary: American Statistical Association Section on Statistics and the Environment (ENVR)	2020

	Treasurer: American Statistical Association Section on Statistics and the Environment (ENVR)	2019
	Associate Editor: <i>Data Science in Science</i> <i>Stat</i>	2021–Current 2023–Current
	Reviewer for the following journals: <i>Annals of Applied Statistics</i> <i>Advances in Statistical Analytics</i> <i>Bayesian Analysis</i> <i>Biometrics</i> <i>Computational Statistics and Data Analysis</i> <i>Environmetrics</i> <i>Environmental and Ecological Statistics</i> <i>Journal of the American Statistical Association</i> <i>Journal of Agricultural, Biological, and Environmental Statistics</i> <i>Journal of Computational and Graphical Statistics</i> <i>Journal of the Royal Statistical Society: Series C</i> <i>Network Science</i> <i>Ecological Monographs</i> <i>Ecology</i> <i>Ecology and Evolution</i> <i>Ecosphere</i> <i>Methods in Ecology and Evolution</i> <i>Movement Ecology</i> <i>Network Science</i> <i>Proceedings of the Royal Society: Series B</i> <i>PLOS ONE</i>	
SOFTWARE	Scharf, H. R. , K. Dinh*, H. Rosales*, A. Rivera* (2025). anipaths: Animation of observed trajectories using spline- or state-space model-based interpolation. R package version 0.10.5.	
OTHER CONFERENCES & WORKSHOPS	Participant in NIMBioS Investigative Workshop <i>Transients in Biological Systems</i> .	2019
	Member of the working group <i>Network Models</i> in the SAMSI 2014-15 Program on Mathematical and Statistical Ecology (ECOL).	2014–2015
	Pan-American Advanced Studies Institute (Buzios, Brazil) Workshop focused on spatiotemporal statistics, with a special emphasis on international collaboration.	2014
	Design and Analysis of Experiments Conference	2012
PROFESSIONAL AFFILIATIONS	International Biometric Society WNAR region	2017–Current

*Student contributor

International Society for Bayesian Analysis	<i>2017–Current</i>
Institute of Mathematical Statistics	<i>2015–Current</i>
American Statistical Association ENVR section	<i>2013–Current</i>